DALHOUSIE UNIVERSITY

# CSCI 5408: Data Management, Warehousing and Analytics

## Assignment: Lab - 03

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## 1) Conceptual Model: ER Diagram using Chen's notations

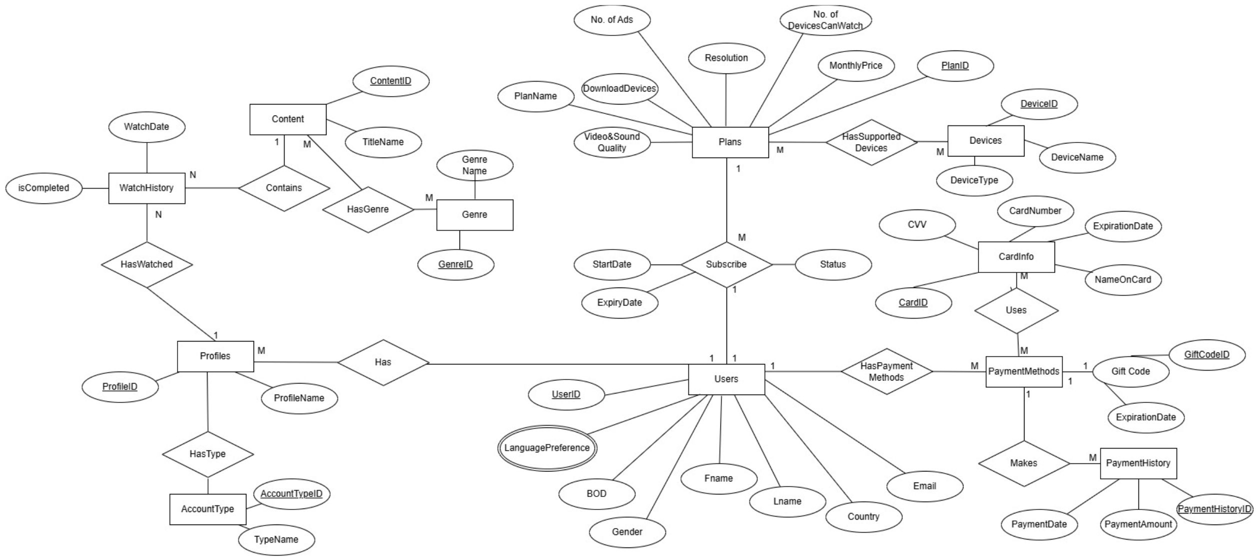
### Key Entities and Attributes

1. Users: It represents individuals registered on the platform with attributes such as **userID, Name** and **Email.**
2. Plans: It defines subscription for users, including attributes like **PlanID, PlanName** and **MontlyPrice.**
3. PaymentMethods: It captures different payment options to users such as cards or gift codes.
4. PaymentHistory: It shows the payment transactions with attributes like **PaymentID**, **Amount** and **PaymentDate**.
5. CardInfo: It shows the card information including **CardID, NameOnCard** and **ExpirationDate.**
6. Devices: It tracks on different devices that users can stream on, such as **DeviceID, DeviceType** and **DeviceName**.
7. Genre: It represents categories of content like **GenreID** and **GenreName.**
8. Content: It describes the movies and TV shows with attributes like **ContentID** and **TitleName.**
9. Profiles: It shows the individual user profiles under a user account with attributes like **ProfileID** and **ProfileName.**
10. AccountType: It specifies the type of profile **for eg. Kids, Adult** using attributes like **AccountTypeID** and **TypeName.**
11. WatchHistory: It records viewing activity for each profile with attributes like **ContentID** and **WatchDate.**

### Relationships

1. Users and Profiles(1:M): A user can have multiple profiles.
2. Profiles and AccountType(M:1): A profile is linked to one account type.
3. Profiles and Watchhistory(M:M): Profiles can watch multiple content items.
4. Users and Plans(M:1): A user has subscribed to one subscription plan.
5. Users and PaymentMethods(1:M): A user can have multiple option for payment.
6. PaymentMethods and CardInfo(1:1): A payment method include one card details.
7. PaymentHistory and PaymentMethods(M:1): A user can have many transactions.
8. Content and Genre(M:M): Many content items belong to multiple genres.
9. Content and WatchHistory(M:M): Content appears in multiple watch histories.

**ER Diagram**




**Explanation**

1. Users can create multiple personalized profiles under a single account.
2. Subscription plans determine the access level for each profile, such as the number of screens or video quality.
3. Payments are managed through various methods like cards or gift codes with the payment history.
4. In the genres, movies and TV shows are categorized so it can be easier for the user which content they watched or love the most.
5. There are two account types like kids and adult ensure appropriate content restrictions, providing a safe and personalized experience for all profiles.

## 2) Logical Model

**Normalizations**

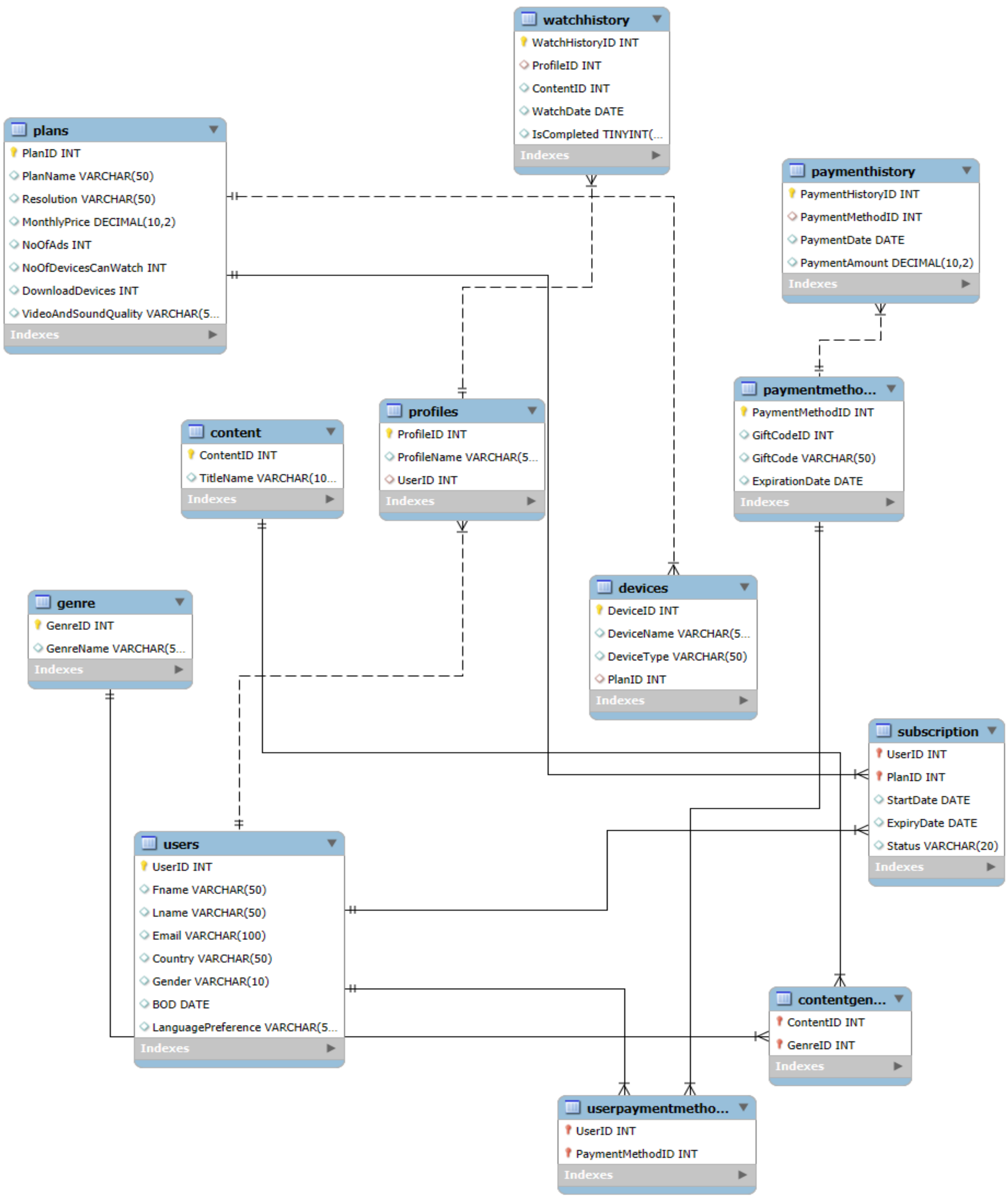
1. 1NF: Already in 1NF because they have atomic attributes and there are no repeating groups or multi-valued attributes.

2. 2NF: It is also in 2NF because all non-prime attributes are fully functionally dependent on the entire primary key, ensuring no partial dependencies.

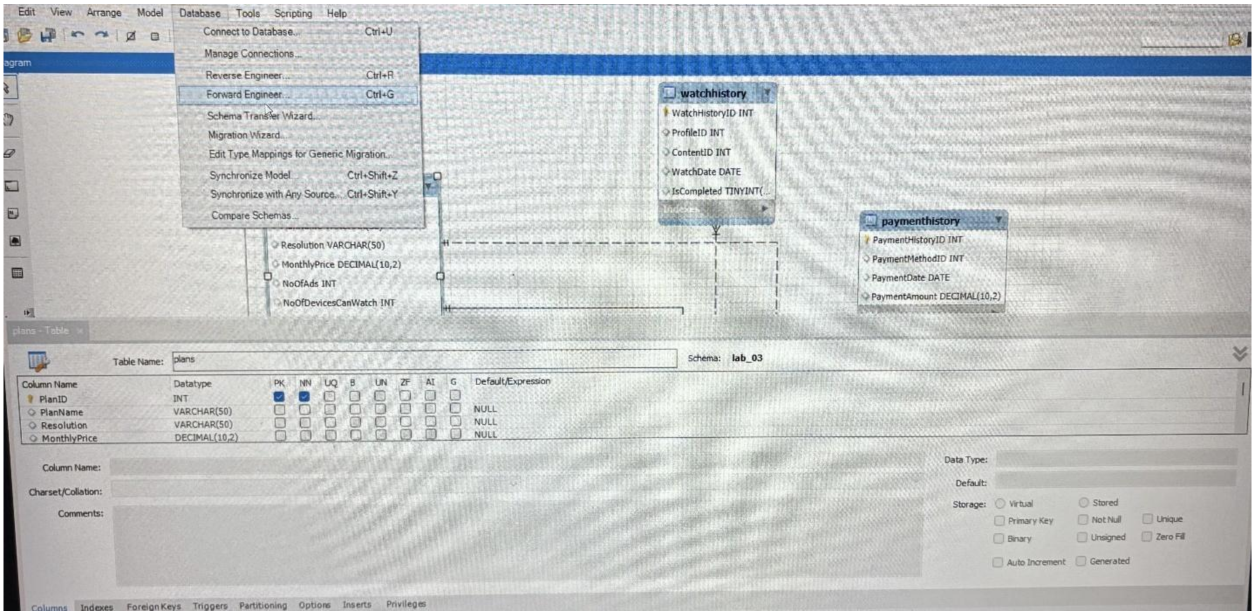
3. 3NF: Already in 3NF as it's already in 2NF and there are no transitive dependencies.

## 3) Physical Model

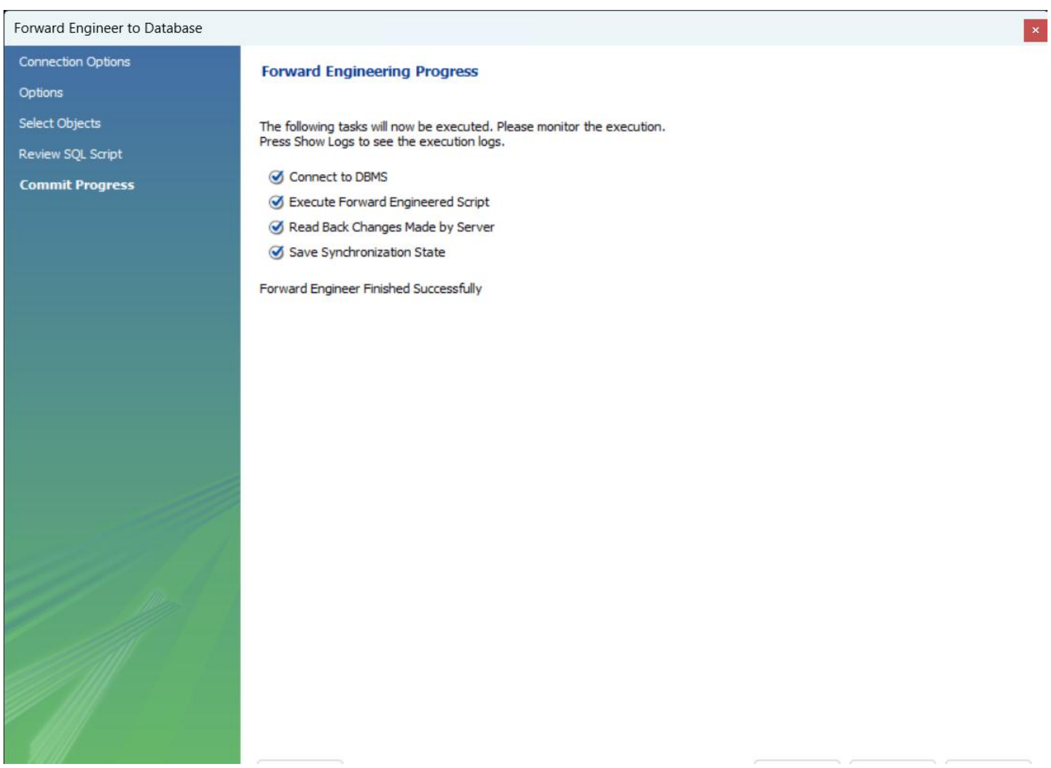
### Crow's Foot Model



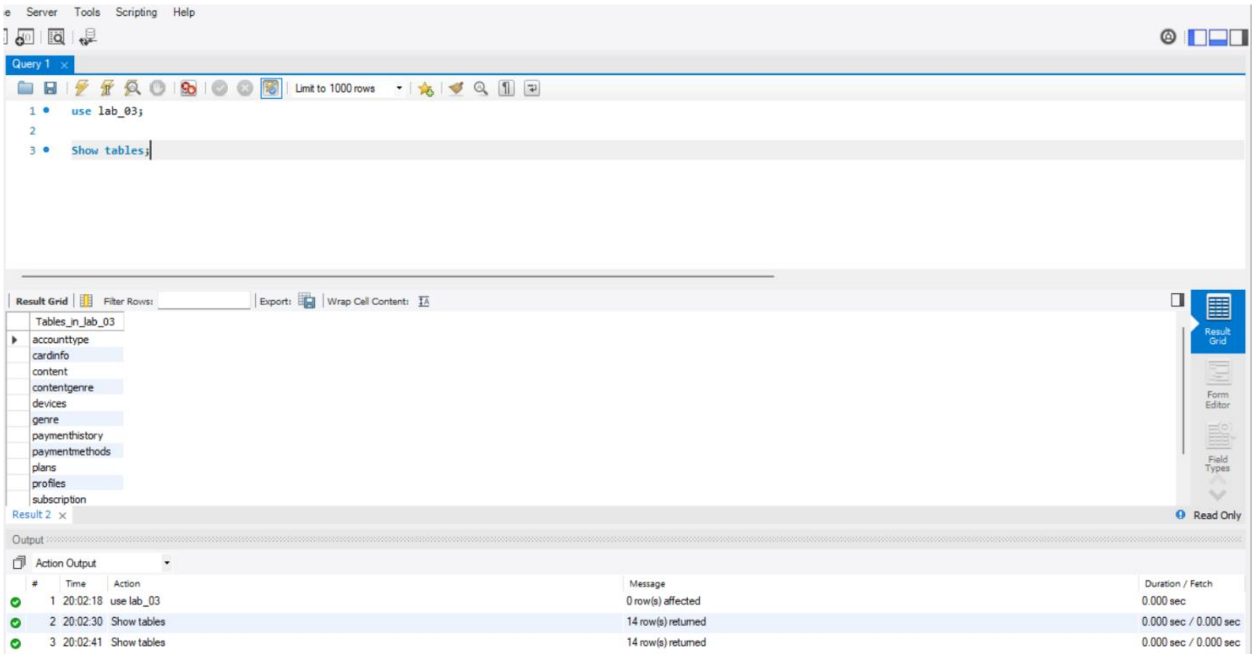
After doing the Crow's Foot Model for the logical model in MYSQL Workbench, I ran forward engineering. Following are the steps:




Selecting Forward Engineer




Forward Engineer Finished Successfully




Showing Tables

**DDL Queries**

-- MySQL Workbench Forward Engineering

SET @OLD_UNIQUE_CHECKS=@@UNIQUE_CHECKS, UNIQUE_CHECKS=0;

SET @OLD_FOREIGN_KEY_CHECKS=@@FOREIGN_KEY_CHECKS, FOREIGN_KEY_CHECKS=0;

SET @OLD_SQL_MODE=@@SQL_MODE, SQL_MODE='ONLY_FULL_GROUP_BY,STRICT_TRANS_TABLES,NO_ZERO_IN_DATE,NO_ZERO_DATE,ERROR_FOR_DIVISION_BY_ZERO,NO_ENGINE_SUBSTITUTION';

-- -----------------------------------------------------

-- Schema mydb

-- -----------------------------------------------------

-- -----------------------------------------------------

```
-- Schema lab3
-- -----------------------------------------------------

-- -----------------------------------------------------
-- Schema lab3
-- -----------------------------------------------------
CREATE SCHEMA IF NOT EXISTS `lab3` ;
-- -----------------------------------------------------
-- Schema NetflixERD
-- -----------------------------------------------------
-- -----------------------------------------------------
-- Schema lab_3
-- -----------------------------------------------------

-- -----------------------------------------------------
-- Schema lab_3
-- -----------------------------------------------------
CREATE SCHEMA IF NOT EXISTS `lab_3` DEFAULT CHARACTER SET utf8mb4
COLLATE utf8mb4_0900_ai_ci ;
-- -----------------------------------------------------
-- Schema lab_03
-- -----------------------------------------------------

-- -----------------------------------------------------
-- Schema lab_03
-- -----------------------------------------------------
CREATE SCHEMA IF NOT EXISTS `lab_03` DEFAULT CHARACTER SET utf8mb4
COLLATE utf8mb4_0900_ai_ci ;
```

```sql
USE `lab3` ;

-- -----------------------------------------------------
-- Table `lab3`.`student`
-- -----------------------------------------------------
CREATE TABLE IF NOT EXISTS `lab3`.`student` (
  `idstudent` INT NOT NULL,
  `name` VARCHAR(45) NOT NULL,
  `dob` DATE NOT NULL,
  `addressl` VARCHAR(300) NOT NULL,
  PRIMARY KEY (`idstudent`))
ENGINE = InnoDB;


-- -----------------------------------------------------
-- Table `lab3`.`University`
-- -----------------------------------------------------
CREATE TABLE IF NOT EXISTS `lab3`.`University` (
  `idUniversity` INT NOT NULL,
  `universityName` VARCHAR(45) NOT NULL,
  `student_universityId` INT NULL,
  PRIMARY KEY (`idUniversity`),
  INDEX `student_universityId_idx` (`student_universityId` ASC) VISIBLE,
  CONSTRAINT `student_universityId`
    FOREIGN KEY (`student_universityId`)
    REFERENCES `lab3`.`student` (`idstudent`)
    ON DELETE NO ACTION
```

```sql
    ON UPDATE NO ACTION)
ENGINE = InnoDB;

USE `lab_3` ;

-- -----------------------------------------------------
-- Table `lab_3`.`users`
-- -----------------------------------------------------
CREATE TABLE IF NOT EXISTS `lab_3`.`users` (
  `UserID` INT NOT NULL AUTO_INCREMENT,
  `Fname` VARCHAR(50) NULL DEFAULT NULL,
  `Lname` VARCHAR(50) NULL DEFAULT NULL,
  `Gender` VARCHAR(10) NULL DEFAULT NULL,
  `BOD` DATE NULL DEFAULT NULL,
  `Age` INT NULL DEFAULT NULL,
  `Email` VARCHAR(100) NULL DEFAULT NULL,
  `Country` VARCHAR(50) NULL DEFAULT NULL,
  PRIMARY KEY (`UserID`),
  UNIQUE INDEX `Email` (`Email` ASC) VISIBLE)
ENGINE = InnoDB
DEFAULT CHARACTER SET = utf8mb4
COLLATE = utf8mb4_0900_ai_ci;


-- -----------------------------------------------------
-- Table `lab_3`.`profiles`
-- -----------------------------------------------------
```

```sql
CREATE TABLE IF NOT EXISTS `lab_3`.`profiles` (
  `ProfileID` INT NOT NULL AUTO_INCREMENT,
  `ProfileName` VARCHAR(50) NULL DEFAULT NULL,
  `LanguagePreference` VARCHAR(50) NULL DEFAULT NULL,
  `UserID` INT NULL DEFAULT NULL,
  PRIMARY KEY (`ProfileID`),
  INDEX `UserID` (`UserID` ASC) VISIBLE,
  CONSTRAINT `profiles_ibfk_1`
    FOREIGN KEY (`UserID`)
    REFERENCES `lab_3`.`users` (`UserID`)
    ON DELETE CASCADE)
ENGINE = InnoDB
DEFAULT CHARACTER SET = utf8mb4
COLLATE = utf8mb4_0900_ai_ci;


-- -----------------------------------------------------
-- Table `lab_3`.`devices`
-- -----------------------------------------------------
CREATE TABLE IF NOT EXISTS `lab_3`.`devices` (
  `DeviceID` INT NOT NULL AUTO_INCREMENT,
  `DeviceType` VARCHAR(50) NULL DEFAULT NULL,
  PRIMARY KEY (`DeviceID`))
ENGINE = InnoDB
DEFAULT CHARACTER SET = utf8mb4
COLLATE = utf8mb4_0900_ai_ci;
```

```sql
-- -----------------------------------------------------
-- Table `lab_3`.`accesses`
-- -----------------------------------------------------
CREATE TABLE IF NOT EXISTS `lab_3`.`accesses` (
  `ProfileID` INT NOT NULL,
  `DeviceID` INT NOT NULL,
  PRIMARY KEY (`ProfileID`, `DeviceID`),
  INDEX `DeviceID` (`DeviceID` ASC) VISIBLE,
  CONSTRAINT `accesses_ibfk_1`
    FOREIGN KEY (`ProfileID`)
    REFERENCES `lab_3`.`profiles` (`ProfileID`)
    ON DELETE CASCADE,
  CONSTRAINT `accesses_ibfk_2`
    FOREIGN KEY (`DeviceID`)
    REFERENCES `lab_3`.`devices` (`DeviceID`)
    ON DELETE CASCADE)
ENGINE = InnoDB
DEFAULT CHARACTER SET = utf8mb4
COLLATE = utf8mb4_0900_ai_ci;


-- -----------------------------------------------------
-- Table `lab_3`.`adultaccount`
-- -----------------------------------------------------
CREATE TABLE IF NOT EXISTS `lab_3`.`adultaccount` (
  `ProfileID` INT NOT NULL,
```

```sql
  PRIMARY KEY (`ProfileID`),
  CONSTRAINT `adultaccount_ibfk_1`
    FOREIGN KEY (`ProfileID`)
    REFERENCES `lab_3`.`profiles` (`ProfileID`)
    ON DELETE CASCADE)
ENGINE = InnoDB
DEFAULT CHARACTER SET = utf8mb4
COLLATE = utf8mb4_0900_ai_ci;


-- -----------------------------------------------------
-- Table `lab_3`.`content`
-- -----------------------------------------------------
CREATE TABLE IF NOT EXISTS `lab_3`.`content` (
  `ContentID` INT NOT NULL AUTO_INCREMENT,
  `TitleName` VARCHAR(100) NULL DEFAULT NULL,
  `Genre` VARCHAR(50) NULL DEFAULT NULL,
  PRIMARY KEY (`ContentID`))
ENGINE = InnoDB
DEFAULT CHARACTER SET = utf8mb4
COLLATE = utf8mb4_0900_ai_ci;


-- -----------------------------------------------------
-- Table `lab_3`.`watchhistory`
-- -----------------------------------------------------
CREATE TABLE IF NOT EXISTS `lab_3`.`watchhistory` (
```

```
  `WatchHistoryID` INT NOT NULL AUTO_INCREMENT,
  `ProfileID` INT NULL DEFAULT NULL,
  `ContentID` INT NULL DEFAULT NULL,
  `WatchDate` DATE NULL DEFAULT NULL,
  `IsCompleted` TINYINT(1) NULL DEFAULT NULL,
  PRIMARY KEY (`WatchHistoryID`),
  INDEX `ProfileID` (`ProfileID` ASC) VISIBLE,
  INDEX `ContentID` (`ContentID` ASC) VISIBLE,
  CONSTRAINT `watchhistory_ibfk_1`
    FOREIGN KEY (`ProfileID`)
    REFERENCES `lab_3`.`profiles` (`ProfileID`)
    ON DELETE CASCADE,
  CONSTRAINT `watchhistory_ibfk_2`
    FOREIGN KEY (`ContentID`)
    REFERENCES `lab_3`.`content` (`ContentID`)
    ON DELETE CASCADE)
ENGINE = InnoDB
DEFAULT CHARACTER SET = utf8mb4
COLLATE = utf8mb4_0900_ai_ci;


-- -----------------------------------------------------
-- Table `lab_3`.`contains`
-- -----------------------------------------------------
CREATE TABLE IF NOT EXISTS `lab_3`.`contains` (
  `WatchHistoryID` INT NOT NULL,
  `ContentID` INT NOT NULL,
```

```
  PRIMARY KEY (`WatchHistoryID`, `ContentID`),
  INDEX `ContentID` (`ContentID` ASC) VISIBLE,
  CONSTRAINT `contains_ibfk_1`
    FOREIGN KEY (`WatchHistoryID`)
    REFERENCES `lab_3`.`watchhistory` (`WatchHistoryID`)
    ON DELETE CASCADE,
  CONSTRAINT `contains_ibfk_2`
    FOREIGN KEY (`ContentID`)
    REFERENCES `lab_3`.`content` (`ContentID`)
    ON DELETE CASCADE)
ENGINE = InnoDB
DEFAULT CHARACTER SET = utf8mb4
COLLATE = utf8mb4_0900_ai_ci;


-- -----------------------------------------------------
-- Table `lab_3`.`giftcode`
-- -----------------------------------------------------
CREATE TABLE IF NOT EXISTS `lab_3`.`giftcode` (
  `GiftCodeID` INT NOT NULL AUTO_INCREMENT,
  `ExpirationDate` DATE NULL DEFAULT NULL,
  PRIMARY KEY (`GiftCodeID`))
ENGINE = InnoDB
DEFAULT CHARACTER SET = utf8mb4
COLLATE = utf8mb4_0900_ai_ci;
```

```sql
-- -----------------------------------------------------
-- Table `lab_3`.`paymentmethods`
-- -----------------------------------------------------
CREATE TABLE IF NOT EXISTS `lab_3`.`paymentmethods` (
  `CardID` INT NOT NULL AUTO_INCREMENT,
  `CardNumber` VARCHAR(16) NULL DEFAULT NULL,
  `NameOnCard` VARCHAR(100) NULL DEFAULT NULL,
  `ExpirationDate` DATE NULL DEFAULT NULL,
  `CVV` VARCHAR(5) NULL DEFAULT NULL,
  PRIMARY KEY (`CardID`),
  UNIQUE INDEX `CardNumber` (`CardNumber` ASC) VISIBLE)
ENGINE = InnoDB
DEFAULT CHARACTER SET = utf8mb4
COLLATE = utf8mb4_0900_ai_ci;


-- -----------------------------------------------------
-- Table `lab_3`.`haspaymentmethods`
-- -----------------------------------------------------
CREATE TABLE IF NOT EXISTS `lab_3`.`haspaymentmethods` (
  `UserID` INT NOT NULL,
  `CardID` INT NOT NULL,
  PRIMARY KEY (`UserID`, `CardID`),
  INDEX `CardID` (`CardID` ASC) VISIBLE,
  CONSTRAINT `haspaymentmethods_ibfk_1`
    FOREIGN KEY (`UserID`)
    REFERENCES `lab_3`.`users` (`UserID`)
```

```
    ON DELETE CASCADE,
  CONSTRAINT `haspaymentmethods_ibfk_2`
    FOREIGN KEY (`CardID`)
    REFERENCES `lab_3`.`paymentmethods` (`CardID`)
    ON DELETE CASCADE)
ENGINE = InnoDB
DEFAULT CHARACTER SET = utf8mb4
COLLATE = utf8mb4_0900_ai_ci;


-- -----------------------------------------------------
-- Table `lab_3`.`haswatched`
-- -----------------------------------------------------
CREATE TABLE IF NOT EXISTS `lab_3`.`haswatched` (
  `ProfileID` INT NOT NULL,
  `ContentID` INT NOT NULL,
  PRIMARY KEY (`ProfileID`, `ContentID`),
  INDEX `ContentID` (`ContentID` ASC) VISIBLE,
  CONSTRAINT `haswatched_ibfk_1`
    FOREIGN KEY (`ProfileID`)
    REFERENCES `lab_3`.`profiles` (`ProfileID`)
    ON DELETE CASCADE,
  CONSTRAINT `haswatched_ibfk_2`
    FOREIGN KEY (`ContentID`)
    REFERENCES `lab_3`.`content` (`ContentID`)
    ON DELETE CASCADE)
ENGINE = InnoDB
```

```sql
DEFAULT CHARACTER SET = utf8mb4
COLLATE = utf8mb4_0900_ai_ci;


-- -----------------------------------------------------
-- Table `lab_3`.`kidsaccount`
-- -----------------------------------------------------
CREATE TABLE IF NOT EXISTS `lab_3`.`kidsaccount` (
  `ProfileID` INT NOT NULL,
  `ContentRestriction` VARCHAR(100) NULL DEFAULT NULL,
  `ParentalControls` VARCHAR(100) NULL DEFAULT NULL,
  PRIMARY KEY (`ProfileID`),
  CONSTRAINT `kidsaccount_ibfk_1`
    FOREIGN KEY (`ProfileID`)
    REFERENCES `lab_3`.`profiles` (`ProfileID`)
    ON DELETE CASCADE)
ENGINE = InnoDB
DEFAULT CHARACTER SET = utf8mb4
COLLATE = utf8mb4_0900_ai_ci;


-- -----------------------------------------------------
-- Table `lab_3`.`paymenthistory`
-- -----------------------------------------------------
CREATE TABLE IF NOT EXISTS `lab_3`.`paymenthistory` (
  `PaymentHistoryID` INT NOT NULL AUTO_INCREMENT,
  `UserID` INT NULL DEFAULT NULL,
```

```sql
  `PaymentAmount` DECIMAL(10,2) NULL DEFAULT NULL,
  `PaymentDate` DATE NULL DEFAULT NULL,
  PRIMARY KEY (`PaymentHistoryID`),
  INDEX `UserID` (`UserID` ASC) VISIBLE,
  CONSTRAINT `paymenthistory_ibfk_1`
    FOREIGN KEY (`UserID`)
    REFERENCES `lab_3`.`users` (`UserID`)
    ON DELETE CASCADE)
ENGINE = InnoDB
DEFAULT CHARACTER SET = utf8mb4
COLLATE = utf8mb4_0900_ai_ci;


-- -----------------------------------------------------
-- Table `lab_3`.`plans`
-- -----------------------------------------------------
CREATE TABLE IF NOT EXISTS `lab_3`.`plans` (
  `PlanID` INT NOT NULL AUTO_INCREMENT,
  `PlanName` VARCHAR(50) NULL DEFAULT NULL,
  `MonthlyPrice` DECIMAL(10,2) NULL DEFAULT NULL,
  `NoOfAds` INT NULL DEFAULT NULL,
  `Resolution` VARCHAR(50) NULL DEFAULT NULL,
  `DownloadDevices` INT NULL DEFAULT NULL,
  `VideoAndSoundQuality` VARCHAR(100) NULL DEFAULT NULL,
  `SupportedDevices` VARCHAR(200) NULL DEFAULT NULL,
  `NoOfDevicesCanWatch` INT NULL DEFAULT NULL,
  PRIMARY KEY (`PlanID`))
```

```
ENGINE = InnoDB
DEFAULT CHARACTER SET = utf8mb4
COLLATE = utf8mb4_0900_ai_ci;


-- -----------------------------------------------------
-- Table `lab_3`.`subscribe`
-- -----------------------------------------------------
CREATE TABLE IF NOT EXISTS `lab_3`.`subscribe` (
  `UserID` INT NOT NULL,
  `PlanID` INT NOT NULL,
  `StartDate` DATE NULL DEFAULT NULL,
  `ExpiryDate` DATE NULL DEFAULT NULL,
  `Status` VARCHAR(50) NULL DEFAULT NULL,
  PRIMARY KEY (`UserID`, `PlanID`),
  INDEX `PlanID` (`PlanID` ASC) VISIBLE,
  CONSTRAINT `subscribe_ibfk_1`
    FOREIGN KEY (`UserID`)
    REFERENCES `lab_3`.`users` (`UserID`)
    ON DELETE CASCADE,
  CONSTRAINT `subscribe_ibfk_2`
    FOREIGN KEY (`PlanID`)
    REFERENCES `lab_3`.`plans` (`PlanID`)
    ON DELETE CASCADE)
ENGINE = InnoDB
DEFAULT CHARACTER SET = utf8mb4
COLLATE = utf8mb4_0900_ai_ci;
```

```
USE `lab_03` ;

-- -----------------------------------------------------
-- Table `lab_03`.`content`
-- -----------------------------------------------------
CREATE TABLE IF NOT EXISTS `lab_03`.`content` (
  `ContentID` INT NOT NULL,
  `TitleName` VARCHAR(100) NULL DEFAULT NULL,
  PRIMARY KEY (`ContentID`))
ENGINE = InnoDB
DEFAULT CHARACTER SET = utf8mb4
COLLATE = utf8mb4_0900_ai_ci;


-- -----------------------------------------------------
-- Table `lab_03`.`genre`
-- -----------------------------------------------------
CREATE TABLE IF NOT EXISTS `lab_03`.`genre` (
  `GenreID` INT NOT NULL,
  `GenreName` VARCHAR(50) NULL DEFAULT NULL,
  PRIMARY KEY (`GenreID`))
ENGINE = InnoDB
DEFAULT CHARACTER SET = utf8mb4
COLLATE = utf8mb4_0900_ai_ci;
```

```sql
-- -----------------------------------------------------
-- Table `lab_03`.`contentgenre`
-- -----------------------------------------------------
CREATE TABLE IF NOT EXISTS `lab_03`.`contentgenre` (
  `ContentID` INT NOT NULL,
  `GenreID` INT NOT NULL,
  PRIMARY KEY (`ContentID`, `GenreID`),
  INDEX `GenreID` (`GenreID` ASC) VISIBLE,
  CONSTRAINT `contentgenre_ibfk_1`
    FOREIGN KEY (`ContentID`)
    REFERENCES `lab_03`.`content` (`ContentID`),
  CONSTRAINT `contentgenre_ibfk_2`
    FOREIGN KEY (`GenreID`)
    REFERENCES `lab_03`.`genre` (`GenreID`))
ENGINE = InnoDB
DEFAULT CHARACTER SET = utf8mb4
COLLATE = utf8mb4_0900_ai_ci;


-- -----------------------------------------------------
-- Table `lab_03`.`plans`
-- -----------------------------------------------------
CREATE TABLE IF NOT EXISTS `lab_03`.`plans` (
  `PlanID` INT NOT NULL,
  `PlanName` VARCHAR(50) NULL DEFAULT NULL,
  `Resolution` VARCHAR(50) NULL DEFAULT NULL,
  `MonthlyPrice` DECIMAL(10,2) NULL DEFAULT NULL,
```

```
  `NoOfAds` INT NULL DEFAULT NULL,
  `NoOfDevicesCanWatch` INT NULL DEFAULT NULL,
  `DownloadDevices` INT NULL DEFAULT NULL,
  `VideoAndSoundQuality` VARCHAR(50) NULL DEFAULT NULL,
  PRIMARY KEY (`PlanID`))
ENGINE = InnoDB
DEFAULT CHARACTER SET = utf8mb4
COLLATE = utf8mb4_0900_ai_ci;


-- -----------------------------------------------------
-- Table `lab_03`.`devices`
-- -----------------------------------------------------
CREATE TABLE IF NOT EXISTS `lab_03`.`devices` (
  `DeviceID` INT NOT NULL,
  `DeviceName` VARCHAR(50) NULL DEFAULT NULL,
  `DeviceType` VARCHAR(50) NULL DEFAULT NULL,
  `PlanID` INT NULL DEFAULT NULL,
  PRIMARY KEY (`DeviceID`),
  INDEX `PlanID` (`PlanID` ASC) VISIBLE,
  CONSTRAINT `devices_ibfk_1`
    FOREIGN KEY (`PlanID`)
    REFERENCES `lab_03`.`plans` (`PlanID`))
ENGINE = InnoDB
DEFAULT CHARACTER SET = utf8mb4
COLLATE = utf8mb4_0900_ai_ci;
```

```
-- -----------------------------------------------------
-- Table `lab_03`.`paymentmethods`
-- -----------------------------------------------------
CREATE TABLE IF NOT EXISTS `lab_03`.`paymentmethods` (
  `PaymentMethodID` INT NOT NULL,
  `GiftCodeID` INT NULL DEFAULT NULL,
  `GiftCode` VARCHAR(50) NULL DEFAULT NULL,
  `ExpirationDate` DATE NULL DEFAULT NULL,
  PRIMARY KEY (`PaymentMethodID`))
ENGINE = InnoDB
DEFAULT CHARACTER SET = utf8mb4
COLLATE = utf8mb4_0900_ai_ci;


-- -----------------------------------------------------
-- Table `lab_03`.`paymenthistory`
-- -----------------------------------------------------
CREATE TABLE IF NOT EXISTS `lab_03`.`paymenthistory` (
  `PaymentHistoryID` INT NOT NULL,
  `PaymentMethodID` INT NULL DEFAULT NULL,
  `PaymentDate` DATE NULL DEFAULT NULL,
  `PaymentAmount` DECIMAL(10,2) NULL DEFAULT NULL,
  PRIMARY KEY (`PaymentHistoryID`),
  INDEX `PaymentMethodID` (`PaymentMethodID` ASC) VISIBLE,
  CONSTRAINT `paymenthistory_ibfk_1`
    FOREIGN KEY (`PaymentMethodID`)
```

```sql
    REFERENCES `lab_03`.`paymentmethods` (`PaymentMethodID`))
ENGINE = InnoDB
DEFAULT CHARACTER SET = utf8mb4
COLLATE = utf8mb4_0900_ai_ci;


-- -----------------------------------------------------
-- Table `lab_03`.`users`
-- -----------------------------------------------------
CREATE TABLE IF NOT EXISTS `lab_03`.`users` (
  `UserID` INT NOT NULL,
  `Fname` VARCHAR(50) NULL DEFAULT NULL,
  `Lname` VARCHAR(50) NULL DEFAULT NULL,
  `Email` VARCHAR(100) NULL DEFAULT NULL,
  `Country` VARCHAR(50) NULL DEFAULT NULL,
  `Gender` VARCHAR(10) NULL DEFAULT NULL,
  `BOD` DATE NULL DEFAULT NULL,
  `LanguagePreference` VARCHAR(50) NULL DEFAULT NULL,
  PRIMARY KEY (`UserID`))
ENGINE = InnoDB
DEFAULT CHARACTER SET = utf8mb4
COLLATE = utf8mb4_0900_ai_ci;


-- -----------------------------------------------------
-- Table `lab_03`.`profiles`
-- -----------------------------------------------------
```

```
CREATE TABLE IF NOT EXISTS `lab_03`.`profiles` (
  `ProfileID` INT NOT NULL,
  `ProfileName` VARCHAR(50) NULL DEFAULT NULL,
  `UserID` INT NULL DEFAULT NULL,
  PRIMARY KEY (`ProfileID`),
  INDEX `UserID` (`UserID` ASC) VISIBLE,
  CONSTRAINT `profiles_ibfk_1`
    FOREIGN KEY (`UserID`)
    REFERENCES `lab_03`.`users` (`UserID`))
ENGINE = InnoDB
DEFAULT CHARACTER SET = utf8mb4
COLLATE = utf8mb4_0900_ai_ci;


-- -----------------------------------------------------
-- Table `lab_03`.`subscription`
-- -----------------------------------------------------
CREATE TABLE IF NOT EXISTS `lab_03`.`subscription` (
  `UserID` INT NOT NULL,
  `PlanID` INT NOT NULL,
  `StartDate` DATE NULL DEFAULT NULL,
  `ExpiryDate` DATE NULL DEFAULT NULL,
  `Status` VARCHAR(20) NULL DEFAULT NULL,
  PRIMARY KEY (`UserID`, `PlanID`),
  INDEX `PlanID` (`PlanID` ASC) VISIBLE,
  CONSTRAINT `subscription_ibfk_1`
    FOREIGN KEY (`UserID`)
```

```sql
    REFERENCES `lab_03`.`users` (`UserID`),
  CONSTRAINT `subscription_ibfk_2`
    FOREIGN KEY (`PlanID`)
    REFERENCES `lab_03`.`plans` (`PlanID`))
ENGINE = InnoDB
DEFAULT CHARACTER SET = utf8mb4
COLLATE = utf8mb4_0900_ai_ci;


-- -----------------------------------------------------
-- Table `lab_03`.`userpaymentmethods`
-- -----------------------------------------------------
CREATE TABLE IF NOT EXISTS `lab_03`.`userpaymentmethods` (
  `UserID` INT NOT NULL,
  `PaymentMethodID` INT NOT NULL,
  PRIMARY KEY (`UserID`, `PaymentMethodID`),
  INDEX `PaymentMethodID` (`PaymentMethodID` ASC) VISIBLE,
  CONSTRAINT `userpaymentmethods_ibfk_1`
    FOREIGN KEY (`UserID`)
    REFERENCES `lab_03`.`users` (`UserID`),
  CONSTRAINT `userpaymentmethods_ibfk_2`
    FOREIGN KEY (`PaymentMethodID`)
    REFERENCES `lab_03`.`paymentmethods` (`PaymentMethodID`))
ENGINE = InnoDB
DEFAULT CHARACTER SET = utf8mb4
COLLATE = utf8mb4_0900_ai_ci;
```

```sql
-- -----------------------------------------------------
-- Table `lab_03`.`watchhistory`
-- -----------------------------------------------------
CREATE TABLE IF NOT EXISTS `lab_03`.`watchhistory` (
  `WatchHistoryID` INT NOT NULL,
  `ProfileID` INT NULL DEFAULT NULL,
  `ContentID` INT NULL DEFAULT NULL,
  `WatchDate` DATE NULL DEFAULT NULL,
  `IsCompleted` TINYINT(1) NULL DEFAULT NULL,
  PRIMARY KEY (`WatchHistoryID`),
  INDEX `ProfileID` (`ProfileID` ASC) VISIBLE,
  CONSTRAINT `watchhistory_ibfk_1`
    FOREIGN KEY (`ProfileID`)
    REFERENCES `lab_03`.`profiles` (`ProfileID`))
ENGINE = InnoDB
DEFAULT CHARACTER SET = utf8mb4
COLLATE = utf8mb4_0900_ai_ci;


SET SQL_MODE=@OLD_SQL_MODE;
SET FOREIGN_KEY_CHECKS=@OLD_FOREIGN_KEY_CHECKS;
SET UNIQUE_CHECKS=@OLD_UNIQUE_CHECKS;
```

## 4) References

[1] “Payment history”, https://www.netflix.com/ca/ [online], Available https://www.netflix.com/ca/ [Accessed: January 24, 2025]

[2] “Free Flowchart Marker and Diagram Software”, https://draw.io/ [Accessed: January 24, 2025]

[3] “MYSQL Workbench”, https://www.mysql.com/products/workbench/ [Accessed: January 24, 2025]